

SparkID

SparkID generates broker authorization from a declared statement of intent, so a Sparkplug estate's access rules are never hand-typed wildcards.

Version 0.1.0 Ignition 8.1.19+ and 8.3 Runs on: Host

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1. Overview

SparkID generates broker authorization for a Sparkplug estate from a declared statement of intent, so a wildcard authorization rule is never hand-typed. An estate needs one identity per edge node, host, replay writer and consumer, each with authorization rules written against the Sparkplug topic grammar; SparkID reads a `.intent` declaration, expands it against that grammar per message type and scope, and emits the broker artifacts. It never writes into a running broker and never opens a network connection; installing the generated artifacts and restarting the broker is a separate, deliberate step.

2. What You Need

- A SparkBridge Host gateway.
- Java 17 if building or running the CLI from source.
- A supported broker target: HiveMQ file-RBAC (`credentials.xml`) or Mosquitto dynamic-security (`--mosquitto`, writes `dynamic-security.json`).

3. Install

SparkID comes with the Host license. Install the module through Config > Modules > Install or Upgrade a Module; no build step is required for the packaged release.

To build the CLI from source:

```
JAVA_HOME=~/.local/opt/jdk-17.0.19+10/Contents/Home ./gradlew fatJar
java -jar build/libs/sparkid-0.1.0-all.jar --help
```

The jar depends on `io.sparkbridge:common` from mavenLocal; publish it first if missing:

```
cd ../sparkbridge && ./gradlew :common:publishToMavenLocal
```

4. License

Soft-licensed; SparkID runs the same whether or not a key is present. License file lookup order: `--license <file>`, the `SPARKBRIDGE_LICENSE` environment variable, then `./sparkbridge-license.key` in the working directory. Run `sparkid license` for full status.

5. Configure

The `.intent` file is a declarative access policy meant for infrastructure-as-code review and diffing. The workflow, start to finish:

1. Validate the written declaration:

```
sparkid validate --intent examples/sparkbridge-dev.intent
```

2. Issue one secret per identity and record it in the ledger:

```
sparkid issue --intent examples/sparkbridge-dev.intent --ledger out/estate.ledger --out-dir out
```

3. Preview the change against the broker's current rules:

```
sparkid diff --intent examples/sparkbridge-dev.intent --current /path/to/credentials.xml
```

4. Generate the reviewed deployment artifacts:

```
sparkid generate --intent examples/sparkbridge-dev.intent --ledger out/estate.ledger --out-dir out --current /path/to/credentials.xml
```

Broker targets	HiveMQ file-RBAC is the only supported target for <code>credentials.xml</code> . Mosquitto dynamic-security is supported with <code>--mosquitto</code> , writing <code>dynamic-security.json</code> .
Write behavior	SparkID never writes into a running broker and never opens a network connection; installing generated artifacts and restarting the broker is a separate, deliberate step.

6. Operate

SparkID is a CLI workflow, not a running service: validate the intent, issue secrets, preview the diff, generate artifacts, then deploy the generated files to the broker by hand and restart it. Re-run `sparkid issue` with the same ledger to rotate credentials inside the overlap window the rotation logic supports.

7. Verify

Run `sparkid diff` against the broker's current `credentials.xml` and confirm the reported changes match intent. Check credential age with `sparkid expiry`. Full acceptance runs are documented in `docs/ACCEPTANCE.md` in the repo.

8. Troubleshoot

Symptom	Cause	Fix
<code>sparkid generate</code> refuses to run	The declared intent would lock out the primary host identity	Review the intent; SparkID refuses a generated policy that would cause host lockout
Broker still enforcing old rules after <code>generate</code>	Generated artifacts were not installed and the broker was not restarted	Copy the generated <code>credentials.xml</code> or <code>dynamic-security.json</code> into place and restart the broker; SparkID never does this for you
Build fails looking for <code>io.sparkbridge:common</code>	Not published to mavenLocal	Run <code>./gradlew :common:publishToMavenLocal</code> in the SparkBridge repo

9. Reference

- CLI commands: `validate`, `issue`, `diff`, `generate`, `expiry`, `license`.
- Broker targets: HiveMQ file-RBAC (default), Mosquitto dynamic-security (`--mosquitto`).
- License file: `sparkbridge-license.key` (working directory, `--license`, or `SPARKBRIDGE_LICENSE`).

10. Support

[SparkID in the docs](#) · [SparkID on the Host product page](#) · [Contact](#) · [GitHub releases](#)